

Self-supervised Phonotactic Representations for Language Identification

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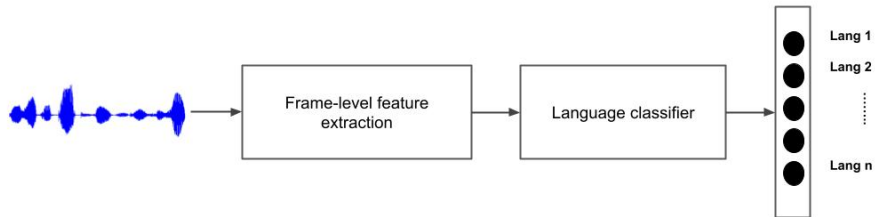
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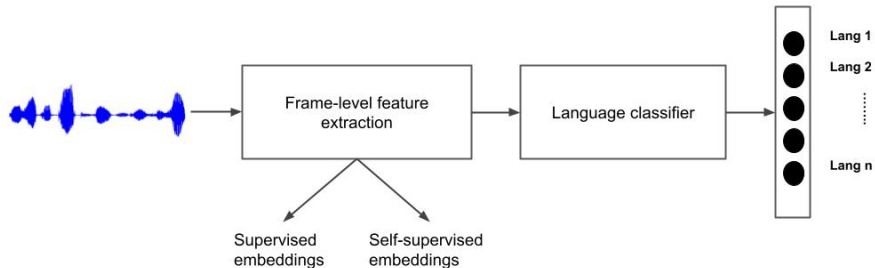
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- Language Identification is task of identifying the language of a spoken utterance.
- Phonotactic constraints characterize the sequence of permissible phoneme structures in a language.
- Short-term spectral analysis is not sufficient to capture them.
- Phonotactic constraints span over hundreds of milli seconds.
- Bottleneck features extracted from a DNN phoneme classifier, have been shown to offer significant improvements.
- Exploring a self-supervised approach to extract phonotactic representations without using transcribed data.

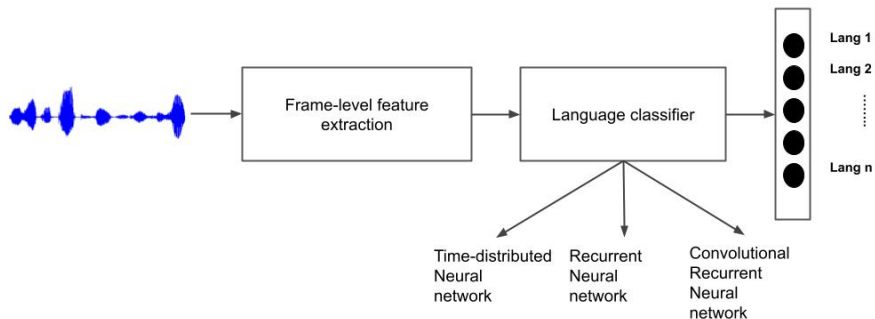
Block diagram for Language identification



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Block diagram for Language identification



Supervised embeddings extraction

- The supervised embeddings are extracted using pre-trained phoneme discriminator model.
- The phoneme discriminator uses neutral resource-rich language Librispeech transcribed data.
- The phoneme discriminator is a TDNN, consists of 17 factorized TDNN layers.
- The pre-final layer (1536) is considered for extracting the bottleneck features.
- The input features for the phoneme discriminator are 40-dimensional MFCC's along with 100-dimensional i-vectors for speaker adaption.

Self-supervised embeddings extraction

- The self-supervised embeddings are extracted using pre-trained wav2vec model.
- Wav2vec uses only neutral resource-rich language Librispeech speech corpus.
- The wav2vec consists of two networks, viz., encoder and context networks.
- The encoder network, with a receptive field of 30 ms, $e : X \rightarrow Z$, maps raw audio X to a latent representation Z .
- The context network, with a receptive field of 810 ms, $a : Z \rightarrow C$ maps a sequence of latent vectors to a contextualized representations
- These 512 dimensional contextualized representations are considered as phonotactic representations.

Supervised embeddings vs Self-supervised embeddings

- Supervised embeddings : Phoneme discriminator
 - The Phoneme discriminator is pre-trained with neutral resource-rich language Librispeech transcribed data.
 - A time-delay neural network architecture with an effective context of 810 ms.
- Self-supervised embeddings : Wav2vec
 - Wav2vec pre-trained with Librispeech without transcribed data.
 - Receptive field of 810 ms is used to capture the long-term contextual information.

Visualization

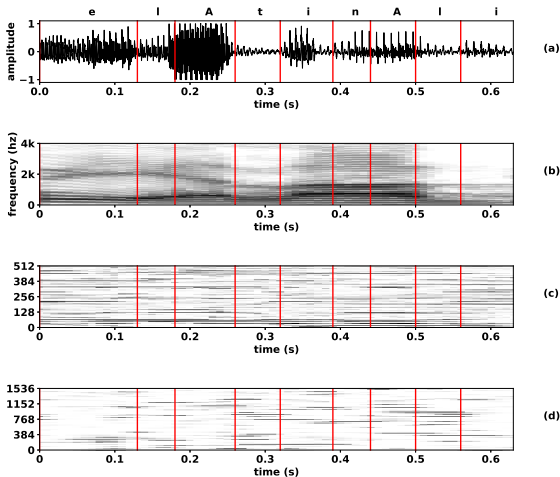


Figure: Visualization of Telugu utterance (a) Waveform with phonemes and their boundaries, (b) Spectrogram, (c) Self-supervised embeddings and (d) Supervised embeddings

Language classifier using utterance-level pooling

- Extracted phonotactic features are need to be distinguishable in the Language space.
- DNNs can learn nonlinear discriminative patterns.
- Time-distributed neural network, RNNs and CRNNs are considered for classification
- All the high-level frame representations do not carry language information
- Attention mechanism is used to pay the attention to particular frames which have discriminative language information

Time-distributed neural network

- Single-layer feed-forward neural network with 128 nodes and ReLu activation
- The weights are shared across the time.
- The number of parameters are 0.1M

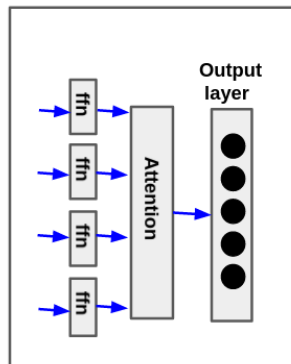


Figure: Architecture for Time-distributed neural network

Recurrent Neural Networks

- The Recurrent neural networks (RNNs) are used to capture the temporal contextual information.
- Three deep Bi directional LSTM layers with residual connections are used
- Each layer is configured with 128 nodes.
- The number of parameters are 0.5M

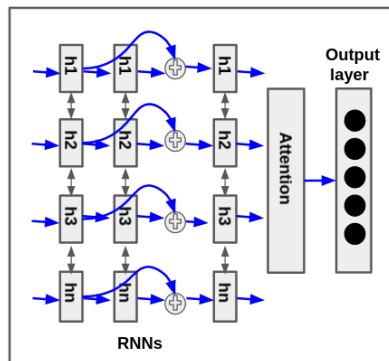


Figure: Architecture for Recurrent Neural Network

Convolutional Recurrent Neural Networks

- We used two 2-D convolutional layers of 32 kernels each with the shapes of (10,10), (3,3) respectively and followed by a max-pooling layer with the shape of (2,2).
- 128 dimensional BLSTM layer is built on the top of the convolutional layers.
- The number of parameters are 0.2M

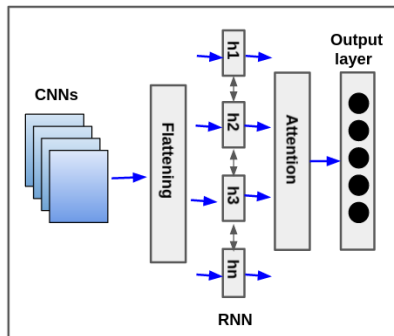


Figure: Architecture for Convolutional Recurrent Neural Network

Proposed architecture

- The LID system consists of 2 stages
 - Extraction of Phonotactic Features
 - Language classifier using utterance-level pooling

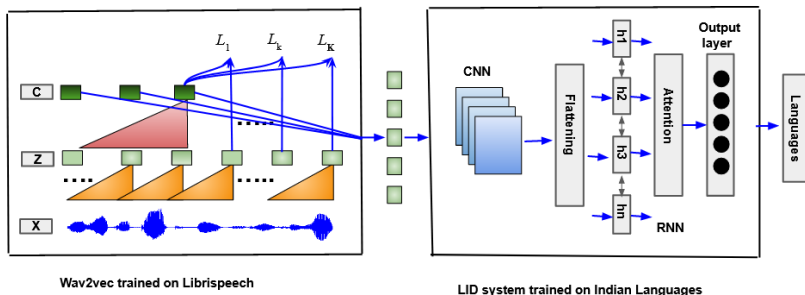


Figure: Proposed architecture for language identification

Dataset Description

- LID system is evaluated on 7 Indian languages viz., Bengali, Hindi, Punjabi, Malayalam, Marathi, Tamil, and Telugu.
- The dataset has been collected from a Youtube channel.
- This channel contains motivational, business and real-life story talks from several speakers.
- The trainset for each language consists of 9 hours from 80 speakers and the testset consists of 1.5 hours from 12 speakers.

Experiments Studies

Features/Networks	Acoustic	Supervised	Self-supervised
Time-distributed	42.15	89.90	91.67
RNNs	67.09	90.1	93.66
CRNNs	70.69	90.73	93.89

Table: LID micro average accuracy of different feature pairs and classifiers

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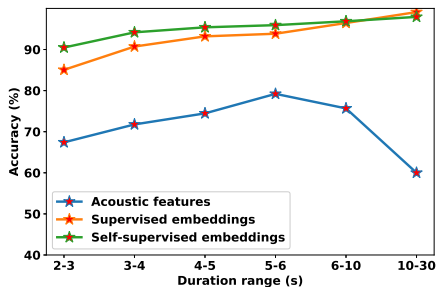
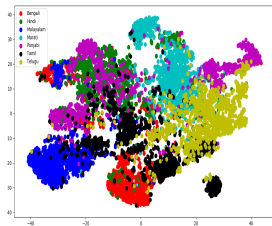
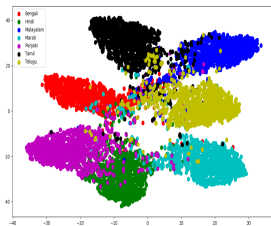


Figure: Effect of test utterance duration on three features with the CRNN classifier

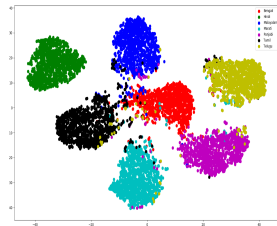
Discriminative embeddings visualization



(a) Acoustic



(b) Supervised



(c) Self-supervised

Figure: t-SNE plots for the embeddings of the CRNN system with various features
color codes : Bengali, Hindi, Malayalam, Marathi, Punjabi, Tamil, Telugu.

Confusion matrix

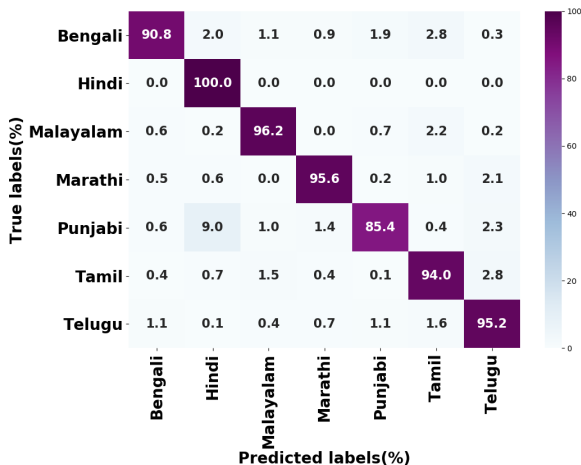


Figure: Confusion matrix of the CRNN system with self-supervised embeddings

Analysis of the Attention scores

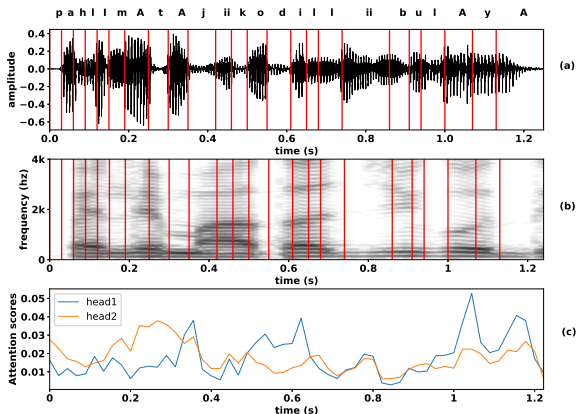


Figure: Visualization of Hindi utterance (a) Waveform with phonemes and their boundaries, (b) Spectrogram and (c) Attention scores

Frequency of occurrence of n-gram

Four-gram sequences/Languages	Hindi	Bengali	Marati	Punjabi	Tamil	Telugu
'm', 'A', 't', 'A'	0.0090	0	0.0050	0	0.0020	0.0024
'l', 'A', 'y', 'A'	0.0077	0	0.0162	0	0	0.0018
'j', 'ii', 'k', 'o'	0.0010	0.0010	0	0	0	0

Table: Frequency of occurrence of four-grams

- The self-supervised embeddings capturing the phonotactic constraints.

- The self-supervised embeddings extracted from the wav2vec network performed better than their supervised counterparts (neutral basis)
- Wav2vec network trained on a neutral language (American English) is good enough to extract language-specific phonotactic constraints.
- Quantifying the capacity of wav2vec embeddings, i.e., the number of target languages they can distinguish when trained with a neutral language.

References



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Thank you